

FGI Digital Ecosystem: Technical Proposal Report

1. Executive Summary: A Vision for Digital Excellence

As Fashion Group International (FGI) approaches nearly a century of influence, the transition from a traditional institution to a digitally native powerhouse requires more than a simple website refresh. It demands a sophisticated, headless architecture designed to provide a premium editorial experience while streamlining complex operational workflows. This proposal outlines the FGI Digital Ecosystem—a future-proof technical foundation architected to scale with FGI's prestige and heritage. The ecosystem is defined by six transformative value pillars:

- **Fashion-Forward Website:** A visually stunning, high-performance platform that serves as FGI's primary brand asset. It utilizes magazine-style layouts and micro-animations to reflect the aesthetics of the fashion, beauty, and creative industries.
- **Unified Membership:** An automated engine managing nine distinct tiers. By removing manual administrative burdens, the system handles multi-step applications, automated renewals, and role-based access for professionals and students alike.
- **Events Engine:** A comprehensive management suite that integrates smart pricing (auto-detecting member status), sponsorship management, and QR-based ticketing to professionalize FGI's event lifecycle.
- **Student Ecosystem:** A dedicated platform for club management and faculty coordination. Most critically, it automates the "graduation-to-professional" pipeline, ensuring FGI retains its next generation of industry leaders.
- **Governance Portal:** A secure HQ workspace designed for the Board of Directors and Regional Directors. It provides a version-controlled document repository with deadline tracking and regional compliance dashboards to ensure operational parity.
- **Enterprise Analytics:** Real-time KPI dashboards that provide the CEO and Board with instant visibility into membership growth, regional participation, and financial health, moving from anecdotal evidence to data-driven strategy.

Key Architectural Decisions

Decision	Rationale
Next.js 15	Industry-leading SSR for SEO and React Server Components for maximum performance on media-heavy pages.

Decision	Rationale
Strapi 5 CMS	A powerful headless CMS that is open-source and self-hosted, eliminating the "per-seat" licensing fees that drain nonprofit budgets.
NestJS API	A type-safe, modular back-end framework ensuring a unified TypeScript stack for rapid development and easier maintenance.
PostgreSQL + Redis	PostgreSQL provides relational integrity for membership data; Redis ensures high-speed session caching and performance.
Stripe Connect	PCI-DSS Level 1 compliant processing for recurring memberships, event ticketing, and automated branded invoicing.
HubSpot CRM	Leverages the Nonprofit Edition to provide enterprise-grade email marketing and segmentation without the overhead of Salesforce.
AWS Cloud	SOC 2 compliant infrastructure designed to scale 10x beyond current traffic without requiring a code re-architecture.

2. Requirements Traceability Matrix

The following matrix maps FGI's specific requirements to our proposed technical solutions, ensuring 100% functional coverage.

RFP Requirement	Proposed Solution	Module
Visually sophisticated design	Next.js + Framer Motion + custom design system	Website
Mobile-responsive, ADA compliant	Responsive CSS + WCAG 2.1 AA automated audit	Website
SEO optimized	Server-Side Rendering (SSR) + structured data	Website
Video/media integration	Cloudinary CDN + embedded high-res video players	Website

RFP Requirement	Proposed Solution	Module
14+ pages + regional directory	Dynamic collections + Map integration	Website
Online membership applications	Multi-step forms with document upload/approval flows	Membership
9 membership levels with RBAC	Middleware-enforced permission matrix	Membership
Payment processing	Stripe Connect + Stripe Subscriptions (auto-renewal)	Membership
Automated reminders & renewals	NestJS cron jobs (90/60/30/7-day intervals)	Membership
Searchable member directory	Elasticsearch with privacy controls	Portal
Role-based dashboards	Custom React dashboards tailored to user role	Portal
Event registration with smart pricing	Membership-aware pricing engine	Events
Student club management	Club registration + Faculty advisor submission workflow	Student
Governance documents	Versioned repository + Deadline tracking	Governance
Targeted email/newsletters	HubSpot CRM integration with bi-directional sync	Comms

3. Recommended Technology Stack

We advocate for a "Full-stack TypeScript" philosophy. By utilizing one language from the database layer to the user interface, we eliminate "translation gaps," reduce development time, and simplify FGI's long-term maintenance requirements.

Front-End

- **Core:** Next.js 15, React 19, TypeScript 5
- **UI/UX:** Framer Motion (micro-animations), Radix UI (accessible primitives), CSS Modules

- **Data Visualization:** Recharts for analytics dashboards

Back-End / API

- **Framework:** NestJS 11, Prisma 6 (ORM), TypeScript 5
- **Logic:** Bull MQ (task scheduling), Passport.js (auth)
- **Automation:** PDFKit (branded invoices), Nodemailer

Content Management

- **System:** Strapi 5 (Headless)
- **Assets:** Cloudinary (Dynamic media hosting and optimization)
- **Access:** Strapi RBAC with REST and GraphQL support

Data Layer

- **Primary DB:** PostgreSQL 16
- **Caching:** Redis 7
- **Search Engine:** Elasticsearch 8 (OpenSearch)

Infrastructure & DevOps

- **Hosting:** AWS ECS Fargate (Serverless container orchestration)
- **Storage/CDN:** AWS S3 + CloudFront
- **DevOps:** GitHub Actions, Terraform (Infrastructure as Code), Docker, Sentry

Why Next.js over WordPress/Drupal?

1. **Pixel-Perfect Design Control:** Fashion-forward sites demand bespoke layouts that template-based themes (WordPress/Drupal) cannot deliver without significant technical debt.
2. **Unrivaled Performance:** SSR ensures Google indexes all content immediately while delivering sub-second page loads.
3. **Modular Reusability:** React components enable consistent patterns across the public site and member portals, reducing future development costs.
4. **Aesthetic High-Resolution:** Automatic image optimization (next/image) serves WebP/AVIF formats—critical for high-fashion imagery where file size often kills site performance.

4. System Architecture & Data Flow

The ecosystem utilizes a headless, domain-driven design. Requests flow through the **AWS CloudFront CDN** to the **Next.js Application Layer**. Content is served via **Strapi 5**, while logic-intensive operations are handled by the **NestJS API Gateway**. This gateway orchestrates data across **PostgreSQL**, **Redis**, and **Elasticsearch**, while syncing with external services like **Stripe** and **HubSpot**.

Role-Based Access Control (RBAC) Hierarchy

The system enforces a precise hierarchy to maintain data security:

- **Level 1: HQ Administrator:** Universal system access and financial oversight.
- **Level 2: Leadership:** Board Members, Regional Directors, and Committee Members.
- **Level 3: Patron:** Enhanced access to exclusive content and VIP features.
- **Level 4: Professional & Corporate:** Standard access to directories and industry reports.
- **Level 5: Student & Emerging Professional:** Base access including student club tools and mentorship matching.

Payment Architecture Workflow

- **Member Portal:** User selects a renewal or event ticket.
- **Checkout (Next.js):** Custom interface calculates "Smart Pricing" (detecting member status).
- **Stripe Connect:** Processes the transaction securely.
- **Webhooks (NestJS):** Receives confirmation and triggers:
- **Update Status:** Membership or registration is immediately activated.
- **Generate Invoice:** Branded PDF created via PDFKit.
- **Send Receipt:** Automated email delivery via AWS SES.

5. CRM Strategy: HubSpot Nonprofit Integration

We recommend HubSpot Nonprofit Edition as the primary CRM. While Salesforce is a common choice, its costs (\$36–\$ 100/user/month) and complexity require dedicated certified administrators. HubSpot provides a more intuitive experience for the FGI team with equal functional parity at a fraction of the cost.

HubSpot CRM Benefits

- ✓ **Free CRM Core:** Unlimited contacts with no per-seat licensing fees.
- ✓ **Nonprofit Discount:** 40% discount on all paid Marketing/Sales hubs.
- ✓ **Email Marketing:** High-end drag-and-drop builder with native A/B testing.
- ✓ **Contact Segmentation:** Automatic grouping by tier, region, and student status.
- ✓ **Workflow Automation:** Managed renewal reminders and welcome sequences.
- ✓ **API Integration:** Robust bi-directional sync via HubSpot API v3.

Integration Flow

1. **Creation:** A new member is created in the NestJS back-end.
2. **Sync:** A webhook triggers an immediate sync to HubSpot CRM, tagging the contact by region and tier.
3. **Onboarding:** HubSpot triggers a 30-day automated welcome email series.

6. Functional Module Implementation

The Public Website (15 Pages)

- **Home:** Hero video, dynamic event calendar, and membership CTAs.
- **Organization:** About FGI (Timeline), Mission/Impact, Leadership (Bios), and Sponsors.
- **Membership:** Tier comparison, benefits, and multi-step application.
- **Community:** Regional Directory (Integrated Map), Student Programs, and News.
- **Resources:** Gated Industry Reports, Scholarships, Education/Webinars, and Contact.

Membership Management Lifecycle

Applications follow a 5-step automated path: **Application** (Form/File Upload) → **Review** (HQ Admin Queue) → **Decision** (Approve/Reject) → **Payment** (Stripe) → **Active** (HubSpot Onboarding).

Member Portal: Role-Based Dashboards

Widget	Student	Prof.	Corp.	Board	Admin
Profile & Events	✓	✓	✓	✓	✓
Member Directory	✓	✓	✓	✓	✓
Renewal Status	✓	✓	✓	✓	✓
Exclusive Content	—	✓	✓	✓	✓
Industry Reports	—	—	✓	✓	✓
Governance Tools	—	—	—	✓	✓

Specialized Engines

- **Events & Registration:** Supports multiple ticket types and sponsor slots. "Smart Pricing" identifies members to apply discounts automatically, while the system generates unique QR-coded PDF tickets for attendance tracking.
- **Student Club Platform:** Includes a faculty advisor registration workflow and a "Mentorship Matching" tool. A "Graduation Trigger" automates the offer of a discounted professional tier upon graduation.
- **Governance Portal:** A repository for version-controlled policies with a **Deadline Tracker** showing status indicators (Submitted, Overdue, Pending) and **Escalation Alerts** for regional compliance.

7. Project Delivery Framework

Development Timeline (11 Months)

- **Phase 1: Discovery & Design (Months 1–1.5):** Interviews, sitemaps, and visual design approval.
- **Phase 2: Core Platform (Months 1.5–3):** Scaffolding, RBAC, and database schema setup.
- **Phase 3: Membership & Portal (Months 3–5):** Approval workflows and Stripe integration.
- **Phase 4: Events & Student Platform (Months 5–7):** Ticketing, smart pricing, and club management.
- **Phase 5: Governance & Communications (Months 7–9):** Repository setup and HubSpot sync.
- **Phase 6: Analytics, QA & Launch (Months 9–11):** Audits, UAT, staff training, and go-live.

Project Team Structure (8.5 FTE)

1. **Project Manager:** Timeline, budget, and risk management.
2. **UX/UI Designer:** Visual design, design system, and accessibility audits.
3. **Sr. Front-End Dev:** Next.js architecture and design system implementation.
4. **Front-End Dev:** Portal dashboards, forms, and directory UI.
5. **Sr. Back-End Dev:** API architecture, authentication, and payments.
6. **Back-End Dev:** Workflows for governance and student platforms.
7. **CMS / Integration:** Strapi setup, HubSpot sync, and data migration.
8. **QA Engineer:** Automated testing and security/accessibility audits.
9. **DevOps Engineer (0.5):** AWS infrastructure, CI/CD, and monitoring.

8. Training, Support, and Operational SLA

Training Plan

Session	Audience	Duration	Topics
CMS Content	HQ/Regional Editors	4 hrs	Pages, media, publishing workflows
Membership Admin	HQ Admin Team	4 hrs	Applications, tiers, reporting

Maintenance & Support

- **Critical (Site Down):** 1 hour response (24/7).
- **High (Feature Broken):** 4 hours response (Business hours).

- **Medium (Minor Issue):** 1 day response.
 - **Low (Enhancement):** 3 days response.
- Monthly Services:** Security patching, daily backups (30-day retention), monthly performance reports, CMS updates, 10 hours of included support, and an annual pentest.

9. Financial Investment Proposal

- **Total Implementation Cost: \$242,500**
- **Annual Maintenance Cost: \$47,900**

Implementation Breakdown

Phase / Service	Duration	Cost (USD)
Development Phases 1–6	11 months	\$229,500
Data Migration	—	\$8,000
Training (7 Sessions)	—	\$5,000
Total Implementation		\$242,500

Annual Maintenance & Hosting

Item	Annual Cost (USD)
AWS Infrastructure	\$9,600
Cloudinary (Media CDN)	\$2,700
Elasticsearch (OpenSearch)	\$3,600
HubSpot CRM (Nonprofit)	\$5,400
Sentry (Error Monitoring)	\$1,200
Domain & SSL	\$200
Monitoring & Security Tools	\$1,200
Support & Maintenance (120 hrs/yr)	\$24,000
Total Annual Cost	\$47,900

Notice of Project Initiation: A two (2) week lead time is required post-contract execution before the project kick-off to allocate resources and provision environments.

Hosting & Services (\$23,900/yr): Direct pass-through utility/licensing fees paid to providers (AWS, HubSpot, Cloudinary, etc.) with no agency markup.

Support & Maintenance (\$24,000/yr): Covers security updates, backup audits, and 120 support hours per year.

Stripe fees (2.9% + \$0.30) are pass-through. HubSpot pricing assumes 40% nonprofit discount.

10. Security, Compliance, and Data Integrity

1. **Encryption:** AES-256 at rest and TLS 1.3 in transit for all data.
2. **Authentication:** NextAuth.js with bcrypt hashing and MFA support.
3. **Authorization:** JWT-based RBAC enforced by NestJS guards.
4. **PCI Compliance:** Stripe Elements integration ensures card data never touches FGI servers.
5. **GDPR:** Consent management with automated data export and deletion APIs.
6. **WAF & DDoS:** AWS WAF (OWASP Top 10) + Shield Standard protection.
7. **Secrets:** AWS Secrets Manager for all API keys and credentials.
8. **Scanning:** Automated vulnerability scanning via Dependabot and Snyk.
9. **Penetration Test:** Annual third-party security pentest.
10. **Backup:** Daily automated backups with 30-day retention and tested recovery.
11. **Audit Logging:** Detailed logs of all admin actions with actor and timestamps.
12. **Privacy:** Cookie consent, profile privacy settings, and vendor DPAs.

11. Conclusion: The Dynamicflow Partnership

Dynamicflow is uniquely positioned to lead FGI's digital transformation:

1. **Full-Stack TypeScript Expertise:** Eliminating translation gaps between front-end and back-end for a cohesive system.
 2. **Nonprofit Specialization:** Deep understanding of tiered memberships, volunteer management, and board governance.
 3. **Fashion Industry Alignment:** Designing to the specific aesthetic standards of fashion, beauty, and creative industries.
 4. **Future-Proof Architecture:** A headless approach that allows for future mobile app expansion without a back-end rebuild.
 5. **Operational Transparency:** Bi-weekly demos and shared project boards to ensure alignment.
 6. **Cost Efficiency:** Use of an open-source stack (Strapi, NestJS) specifically to eliminate recurring enterprise CMS licensing fees.
- Proposal Validity:** 90 days from June 25, 2026. **Contact:** dynamicflowit.com